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Types of Passing Parameters

→ cplusplus.com ←

1 Passing by Value

Pass by Ref. ①

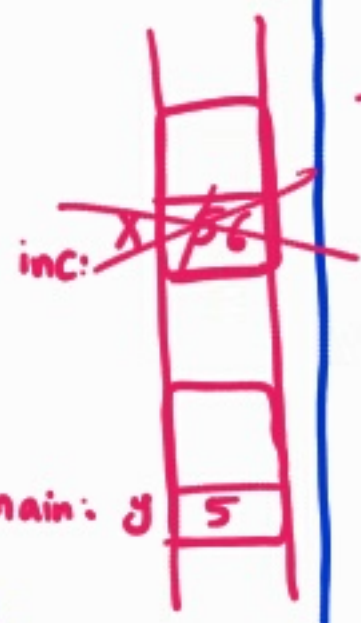
مش كود

```
void inc(int x)
{
    x = x + 1;
}

int main()
{
    int y = 5;
    inc(y);
    cout << y;
}
//5
```

=y

Memory



2 Passing by Pointer

مش كود

```
void inc(int *p)
{
    *p = *p + 1;
}

int main()
{
    int y = 5;
    inc(&y);
    cout << y;
}
//6
```

=&y

3 Passing by Ref.

alias

Pass by Ref ②

مش كود

```
void inc(int &x)
{
    x = x + 1;
}

int main()
{
    int y = 5;
    inc(y);
    cout << y;
}
//6
```

=y

inc: x

main: y

alias

new name

```
int a = 5;
int &b = a;
b = 6;
cout << a;
//6
```

Passing by Pointer/Ref:

When:

- more than 1 output
- Large Input Size (Structs)

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Example:

SafeDivide : divides 2 numbers if denominator != 0, otherwise, returns false.

i/p: x, y

o/p: ① bool (true, false)

② result

Prototype ⇒ bool SafeDivide(int x, int y, int & result)

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Example: * SafeDivide : divides 2 numbers if denominator != 0, otherwise, returns false.

```
bool SafeDivide(int x, int y, float & result)
{
    → if (y != 0)
      {
          → result = x/y;   (float)
          → return true;
      }
    → return false;
}

int main()
{
    int a, b;
    float c;
    cout << "please enter 2 numbers" << endl;
    cin >> a >> b;
    bool s = SafeDivide(a, b, c);
    if (s == true)
        cout << "Successful, Result= " << c << endl;
    else
        cout << " Not Successful" << endl;
}
```

integer division

**

has to be float (video غلطة في ال)

if (s)

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الزتونة :D

Assume: x is int
ptr is int*

Operator (&)

address of operator

&x : pointer

int &x : var

alias

Operator (*)

dereference operator

بيوصلك

*ptr : var

int *ptr : pointer

② ①
*p++; ✗

(*p)++;
① ②

Exercise

one func.

Calculator:

Takes 2 numbers returns their addition, Subtraction and mult.